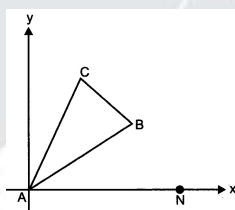


SCQ (Single Correct Type) :

- The minimum value of $\frac{r_1 r_2 r_3}{r^3}$ in a triangle is (symbols have their usual meaning)
(A) 1 (B) 3 (C) 8 (D) 27
- Find the radius of the circle escribed to the triangle ABC (Shown in the figure below) on the side BC if $\angle NAB = 30^\circ$; $\angle BAC = 30^\circ$; $AB = AC = 5$.



- (A) $\frac{(10\sqrt{2} + 5\sqrt{3} - 5)(2 - \sqrt{3})}{2\sqrt{2}}$ (B) $\frac{(10\sqrt{2} + 5\sqrt{3} + 5)(2 - \sqrt{3})}{2\sqrt{2}}$
- (C) $\frac{(10\sqrt{2} + 5\sqrt{3} - 5)(2 + \sqrt{3})}{2\sqrt{2}}$ (D) $\frac{(10\sqrt{2} + 5\sqrt{3} + 1)(\sqrt{3} - 1)}{2\sqrt{2}}$
- Let $\det A = \begin{vmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ and if $(l - m)^2 + (p - q)^2 = 9$, $(m - n)^2 + (q - r)^2 = 16$, $(n - l)^2 + (r - p)^2 = 25$, then the value of $(\det. A)^2$ equals:
(A) 36 (B) 100 (C) 144 (D) 169
 - Number of points of discontinuity of $f(x) = \left\{ \frac{x}{5} \right\} + \left[\frac{x}{2} \right]$ in $x \in [0, 100]$ is/are (where $[\cdot]$ denotes greatest integer function and $\{\cdot\}$ denotes fractional part function)
(A) 50 (B) 51 (C) 52 (D) 61

MCQ (One or more than one correct) :

- Consider right angle triangles of integer side lengths whose product of leg lengths is equal to three times the perimeter, then
(A) Number of such non-congruent triangles is 3
(B) Number of such non-congruent triangles is 4
(C) Maximum possible area of such a triangle is 84 sq. units
(D) Maximum possible area of such a triangle is 81 sq. units

6. 3 circle of radii 1, 2 and 3 and centres at A, B and C respectively, touch each other. Another circle whose centre is P touches all these 3 circles externally. and has radius r. Also $\angle PAB = \theta$ & $\angle PAC = \alpha$.

(A) $\cos \theta = \frac{3-r}{3(1+r)}$ (B) $\cos \alpha = \frac{2-r}{2(1+r)}$ (C) $r = \frac{6}{23}$ (D) $r = \frac{6}{\sqrt{23}}$

7. The value of θ lying between $\frac{-\pi}{4}$ and $\frac{\pi}{2}$ and $0 \leq A \leq \frac{\pi}{2}$ and satisfying the equation

$$\begin{vmatrix} 1 + \sin^2 A & \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & 1 + \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & \cos^2 A & 1 + 2 \sin 4\theta \end{vmatrix} = 0 \text{ are.}$$

(A) $A = \frac{\pi}{4}, \theta = -\frac{\pi}{8}$ (B) $A = \frac{3\pi}{8} = \theta$ (C) $A = \frac{3\pi}{5}, \theta = -\frac{\pi}{8}$ (D) $A = \frac{\pi}{6}, \theta = \frac{3\pi}{8}$

8. Let a, b and c be three real numbers satisfying $[abc] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [0 \ 0 \ 0] \dots (E)$

(A) Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$, If $a = 2$ with b and c satisfying, (E), then the value of

$\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to -2

(B) Let ω be a solution of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$, If $a = 2$ with b and c satisfying, (E), then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to -3

(C) Let $b = 6$ with a and c satisfying (E). If α and β are the roots of the quadratic equation $ax^2 + bx + c =$

0, then $\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n$ is 7

(D) Let $b = 6$ with a and c satisfying (E). If α and β are the roots of the quadratic equation $ax^2 + bx + c =$

0, then $\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n$ is $\frac{6}{7}$

9. Let $D(x) = \begin{vmatrix} x^2 + 4x - 3 & 2x + 4 & 13 \\ 2x^2 + 5x - 9 & 4x + 5 & 26 \\ 8x^2 - 6x + 1 & 16x - 6 & 104 \end{vmatrix} = \alpha x^3 + \beta x^2 + \gamma x + \delta$ then :

(A) $\alpha + \beta = 0$ (B) $\beta + \gamma = 0$ (C) $\alpha + \beta + \gamma + \delta = 0$ (D) $\alpha + \beta + \gamma = 0$

10. If the system of equations

$$ax + y + 2z = 0$$

$$x + 2y + z = b$$

$$2x + y + az = 0$$

has no solution then (a + b) can be equals to :

(A) -1 (B) 2 (C) 3 (D) 4

11. Let $[x]$ denote the greatest integer less than or equal to x . If $f(x) = [x \sin \pi x]$, then $f(x)$ is :
- (A) continuous at $x = 0$ (B) continuous in $(-1, 0)$
 (C) differentiable at $x = 1$ (D) differentiable in $(-1, 1)$
12. $f(x) = \begin{cases} 2(x+1) & ; \quad x \leq -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1, \text{ then :} \\ |||x|-1|-1| & ; \quad x \geq 1 \end{cases}$
- (A) $f(x)$ is non-differentiable at exactly three points
 (B) $f(x)$ is continuous in $(-\infty, 1]$
 (C) $f(x)$ is differentiable in $(-\infty, -1)$
 (D) $f(x)$ is finite type of discontinuity at $x = 1$, but continuous at $x = -1$
13. Let $f(x) = (x+1)(x+2)(x+3) \dots (x+100)$ and $g(x) = f(x)f''(x) - (f'(x))^2$. Let n be the number of real roots of $g(x) = 0$, then :
- (A) $n < 2$ (B) $n > 2$ (C) $n < 100$ (D) $n > 100$
14. $f(x)$ is differentiable function satisfying the relationship $f^2(x) + f^2(y) + 2(xy - 1) = f^2(x+y) \quad \forall x, y \in \mathbb{R}$. Also $f(x) > 0 \quad \forall x \in \mathbb{R}$ and $f(\sqrt{2}) = 2$. Then which of the following statement(s) is/are correct about $f(x)$?
- (A) $[f(3)] = 3$ ($[\cdot]$ denotes greatest integer function) (B) $f(\sqrt{7}) = 3$
 (C) $f(x)$ is even (D) $f'(0) = 0$

Numerical Based Questions :

15. Given that for $a, b, c, d \in \mathbb{R}$, if $a \sec(200^\circ) - c \tan(200^\circ) = d$ and $b \sec(200^\circ) + d \tan(200^\circ) = c$, then find the value of $\left(\frac{a^2 + b^2 + c^2 + d^2}{bd - ac} \right) \sin 20^\circ$.
16. If $\sum_{n=0}^{\infty} 2 \cot^{-1} \left(\frac{n^2 + n + 4}{2} \right) = k\pi$, then find the value of k .
17. In $\triangle ABC$, if r and R are the roots of the equation $x^3 + x^2(1-3p) + (2p^2-3p)x + 2p^2 = 0$ (p is a positive real number) and $c = 6\sqrt{3}$, then find the value of p .

Paragraph Type Question :

Paragraph for Question Nos. 18 to 20

Consider f, g and h be three real valued function defined on $\mathbb{R}$.

Let $f(x) = \sin 3x + \cos x$, $g(x) = \cos 3x + \sin x$ and $h(x) = f^2(x) + g^2(x)$

18. The length of a longest interval in which the function $y = h(x)$ is increasing, is :
- (A) $\frac{\pi}{8}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

19. General solution of the equation $h(x) = 4$, is :

- (A) $(4n+1)\frac{\pi}{8}$ (B) $(8n+1)\frac{\pi}{8}$ (C) $(2n+1)\frac{\pi}{4}$ (D) $(7n+1)\frac{\pi}{4}$

[where $n \in \mathbb{I}$]

20. Number of point(s) where the graphs of the two function, $y = f(x)$ and $y = g(x)$ intersects in $[0, \pi]$, is :

- (A) 2 (B) 3 (C) 4 (D) 5



ANSWERKEY OF CAPS – 10+

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|-----------|------------|----------|-----------|-----------|-----------|
| 1. (D) | 2. (A) | 3. (C) | 4. (A) | 5. (AC) | 6. (ABC) |
| 7. (ABCD) | 8. (AC) | 9. (ABD) | 10. (BCD) | 11. (ABD) | 12. (ACD) |
| 13. (AC) | 14. (ABCD) | 15. 2 | 16. 1 | 17. 3 | 18. (B) |
| 19. (A) | 20. (C) | | | | |